

# INTERNETWORKING

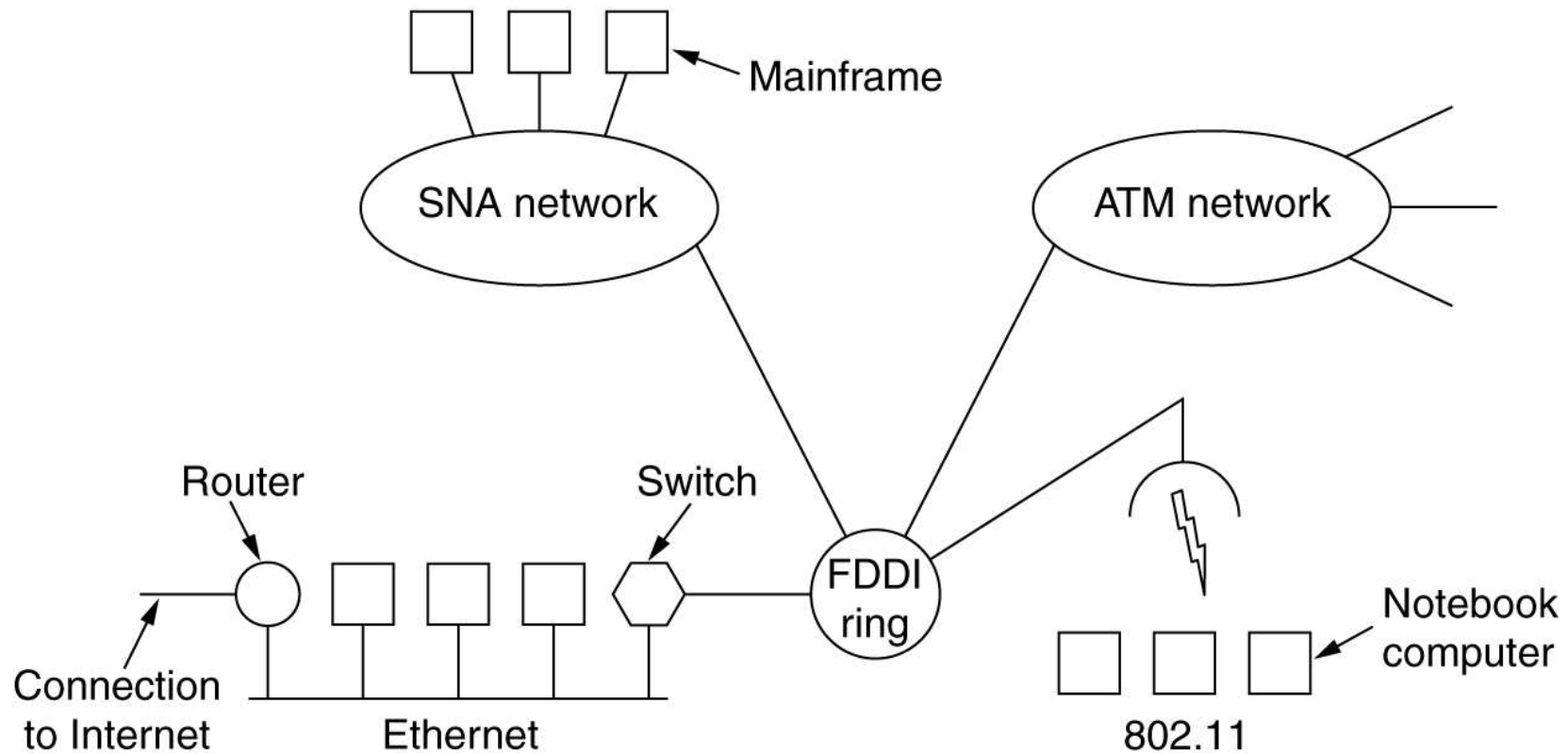
- How Networks Differ
- How Networks Can Be Connected
- Tunneling
- Internetwork Routing
- Packet Fragmentation

# Internetworking: Introduction

- A variety of different networks (and thus protocols) will be around
  - The installed base of different networks is large and growing.
  - As computers and networks get cheaper, the place where decisions get made moves downward.
  - Different networks have radically different technology, so it should not be surprising that as new hardware developments occur, new software will be created to fit the new hardware.

# Internetworking: Introduction

A collection of interconnected networks.



# Internetworking: Introduction

- The interconnection of different networks
  - LAN-LAN: A computer scientist downloading a file to engineering.
  - LAN-WAN: A computer scientist sending mail to a distant physicist.
  - WAN-WAN: Two poets exchanging sonnets.
  - LAN-WAN-LAN: Engineers at different universities communicating.

# Internetworking: How Networks Differ

Some of the many ways networks can differ

| Item               | Some Possibilities                            |
|--------------------|-----------------------------------------------|
| Service offered    | Connectionless versus connection oriented     |
| Addressing         | Different sizes, flat or hierarchical         |
| Broadcasting       | Present or absent (also multicast)            |
| Packet size        | Every network has its own maximum             |
| Ordering           | Ordered and unordered delivery                |
| Quality of service | Present or absent; many different kinds       |
| Reliability        | Different levels of loss                      |
| Security           | Privacy rules, encryption, etc.               |
| Parameters         | Different timeouts, flow specifications, etc. |
| Accounting         | By connect time, packet, byte, or not at all  |

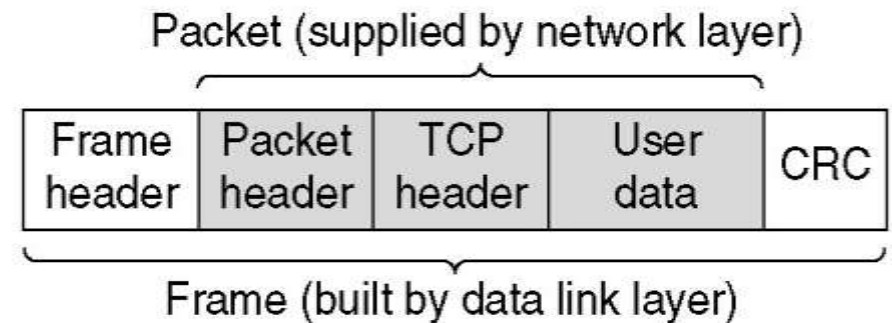
# Internetworking: How Networks Can Be Connected

(a) Which device is in which layer.

(b) Frames, packets, and headers.

|                   |                     |
|-------------------|---------------------|
| Application layer | Application gateway |
| Transport layer   | Transport gateway   |
| Network layer     | Router              |
| Data link layer   | Bridge, switch      |
| Physical layer    | Repeater, hub       |

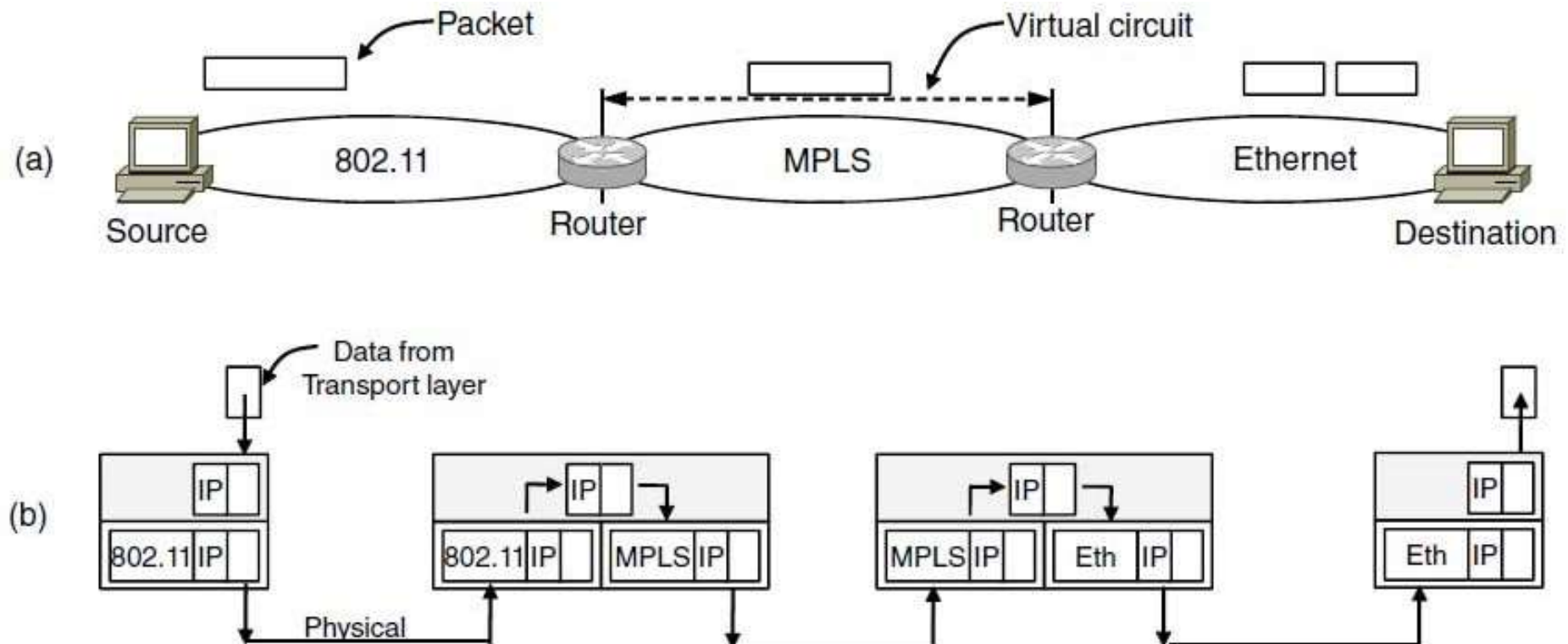
(a)



(b)

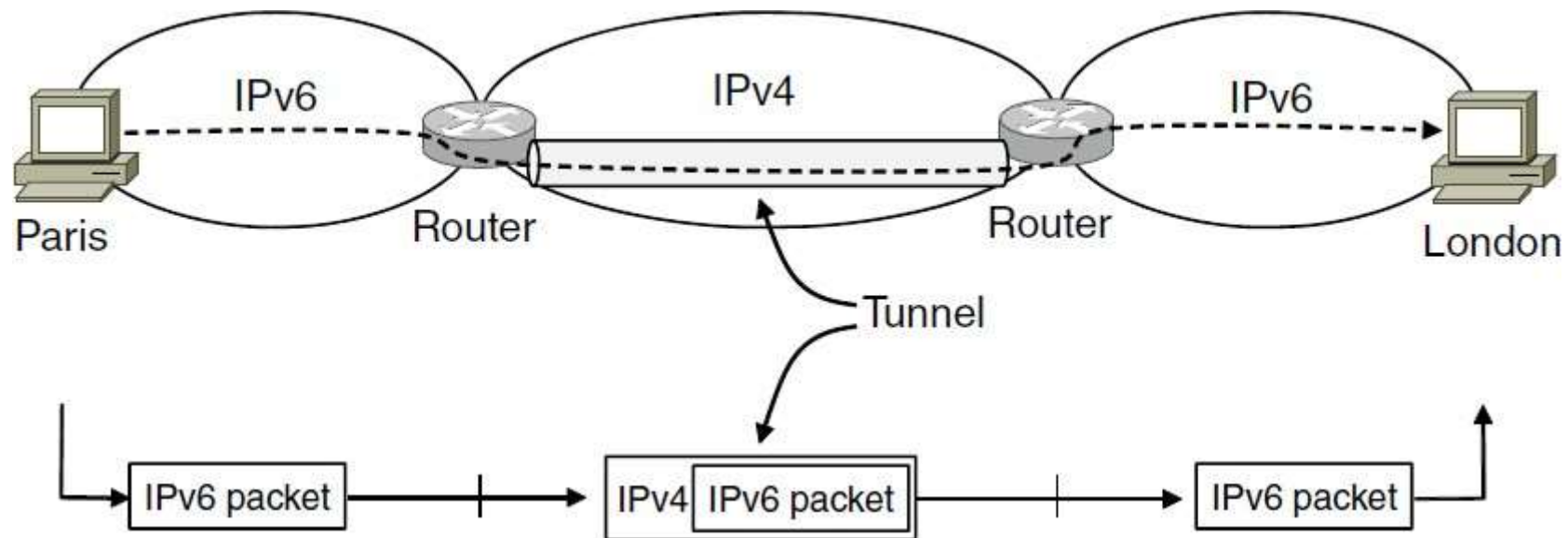
# Internetworking: How Networks Can Be Connected

- (a) A packet crossing different networks.
- (b) Network and link layer protocol processing.



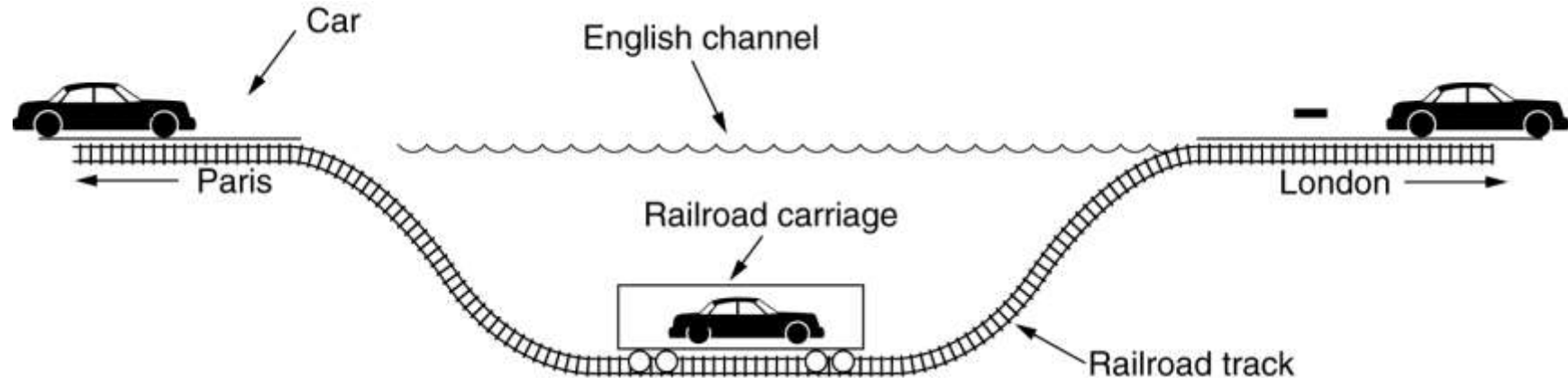
# Internetworking: Tunneling

Tunneling a packet from Paris to London.



# Internetworking: Tunneling

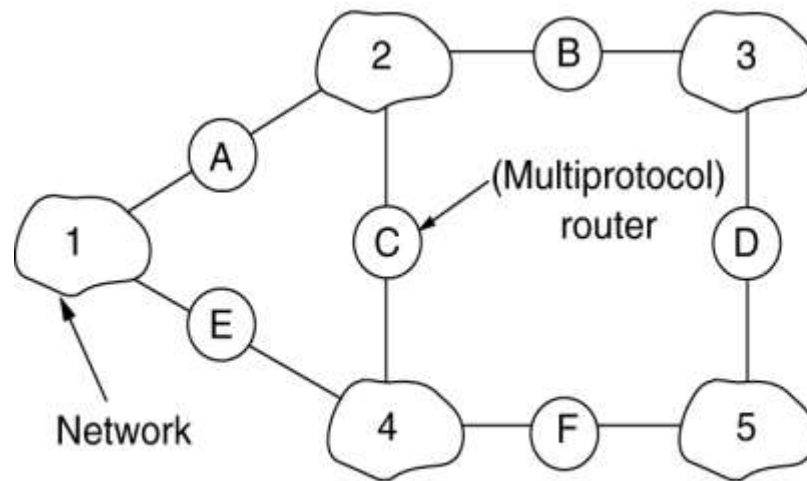
Tunneling a car from France to England.



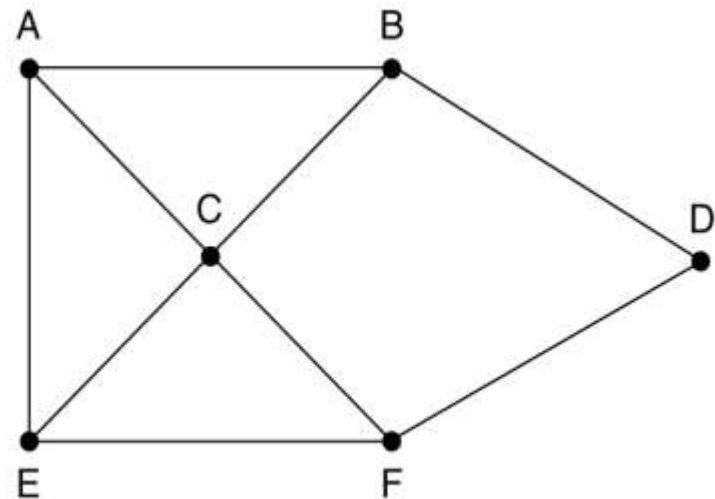
# Internetworking: Internetwork Routing

(a) An internetwork.

(b) A graph of the internetwork.



(a)



(b)

# Internetworking: Internetwork Routing

- Two-level routing
  - Internet routing: Exterior gateway protocol
  - Intranet routing: Interior gateway protocol
- To route a packet
  - A typical internet packet starts out on its LAN addressed to the local multiprotocol router.
  - After it gets there, the network layer code decides which multiprotocol router to forward the packet to, using its own routing tables.
    - Direct forwarding using native network protocol
    - Tunneling using the intervening network protocol
  - This process repeats until the packet reaches the destination network.

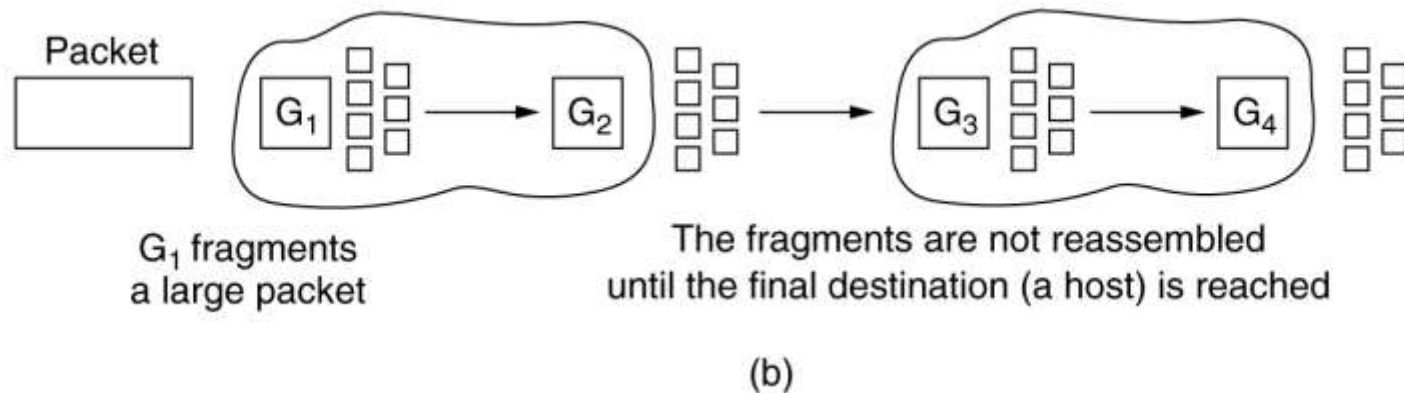
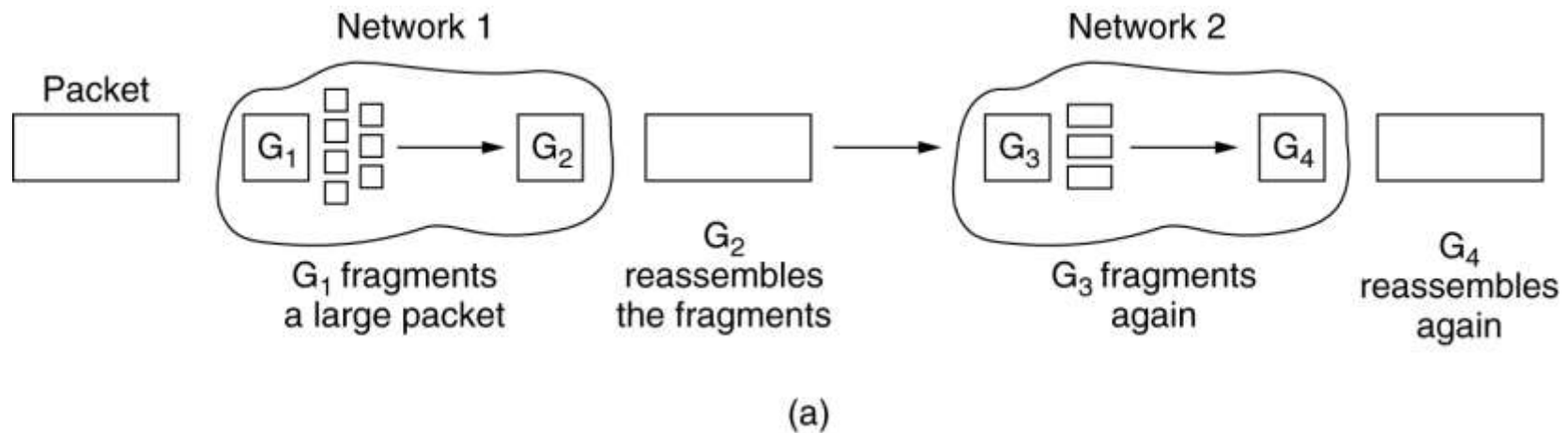
# Internetworking: Fragmentation

- Each network imposes some maximum size on its packets. These limits have various causes, among them:
  - Hardware (e.g., the width of a TDM transmission slot).
  - Operating system (e.g., all buffers are 512 bytes).
  - Protocols (e.g., the number of bits in the packet length field).
  - Compliance with some (inter)national standard.
  - Desire to reduce error induced retransmissions to some level.
  - Desire to prevent one packet from occupying the channel too long.

# Internetworking: Fragmentation

(a) Transparent fragmentation. (ATM)

(b) Nontransparent fragmentation. (IP)



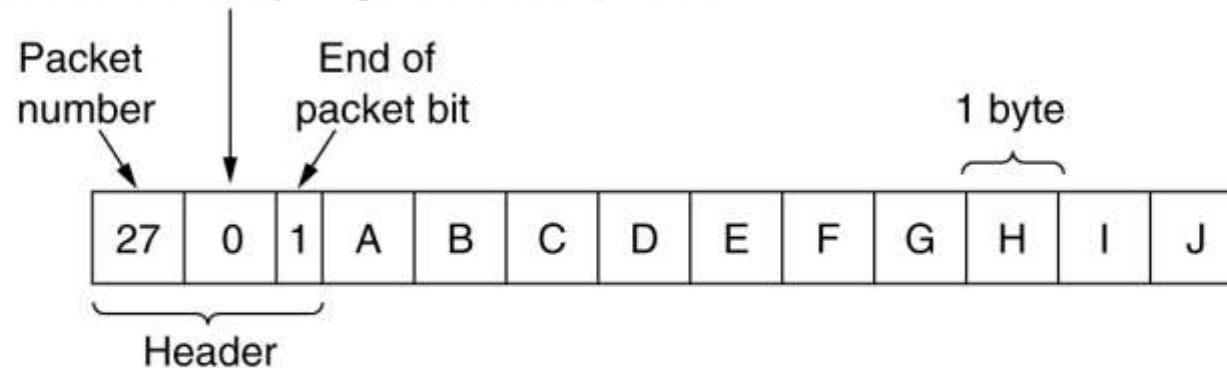
# Internetworking: Fragmentation

- When a packet is fragmented, the fragments must be numbered in such a way that the original data stream can be reconstructed.
  - To define an elementary fragment size small enough that the elementary fragment can pass through every network.
  - When a packet is fragmented, all the pieces are multiple of the elementary fragment size.
  - The internet header provide
    - an original packet number and
    - the number of the (first) elementary fragment contained in the packet and
    - a bit indicating the last piece of the original packet.

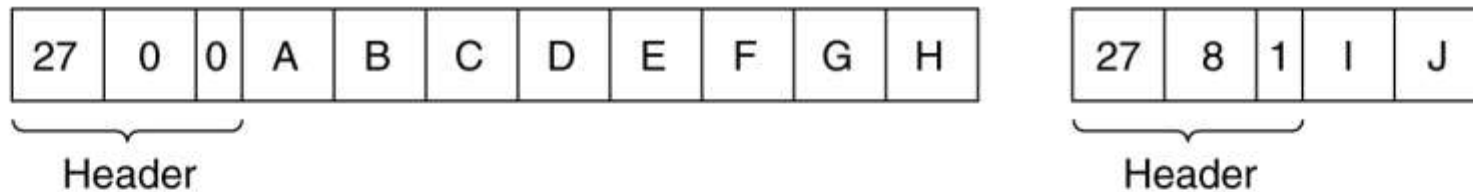
# Internetworking: Fragmentation

Fragmentation when the elementary data size is 1 byte.

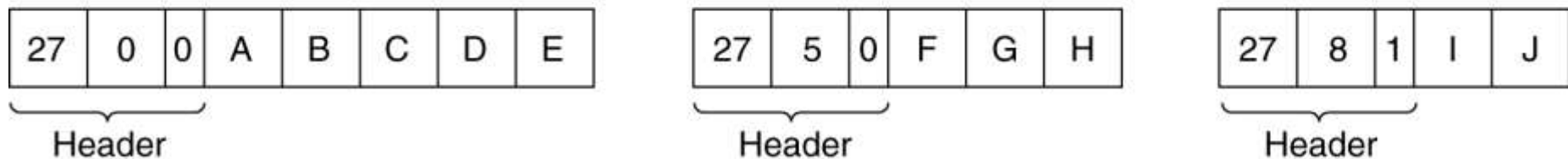
Number of the first elementary fragment in this packet



(a)



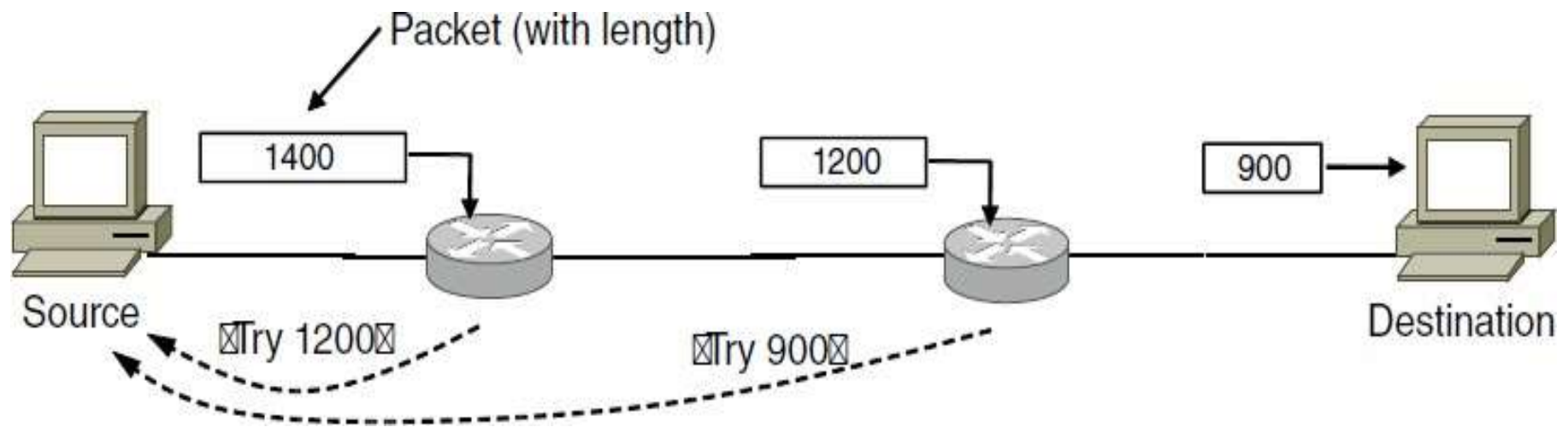
(b)



(c)

# Internetworking: Fragmentation

## Path MTU Discovery



# THE NETWORK LAYER IN THE INTERNET

- The IPv4 Protocol
- IP Addresses
- The IPv6 Protocol
- Internet Control Protocols
- OSPF – The Interior Gateway Routing Protocol
- BGP – The Exterior Gateway Routing Protocol

# The Network Layer in the Internet:

## Top 10 principles for the Internet

1. Make sure it works.
2. Keep it simple.
3. Make clear choices.
4. Exploit modularity.
5. Expect heterogeneity.

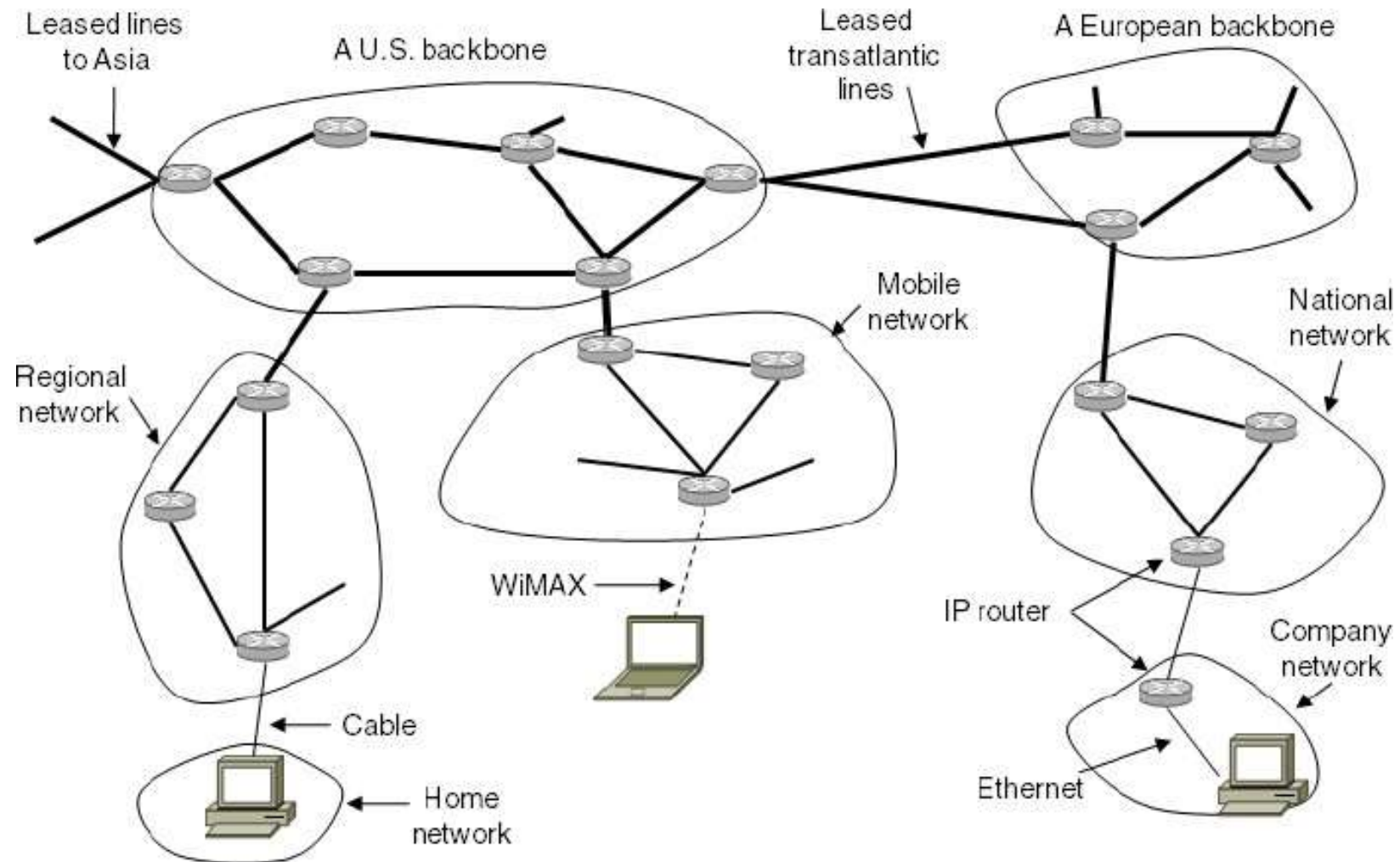
# The Network Layer in the Internet:

## Top 10 principles for the Internet

6. Avoid static options and parameters.
7. Look for a good design; it need not be perfect.
8. Be strict when sending and tolerant when receiving.
9. Think about scalability.
10. Consider performance and cost.

# The Network Layer in the Internet:

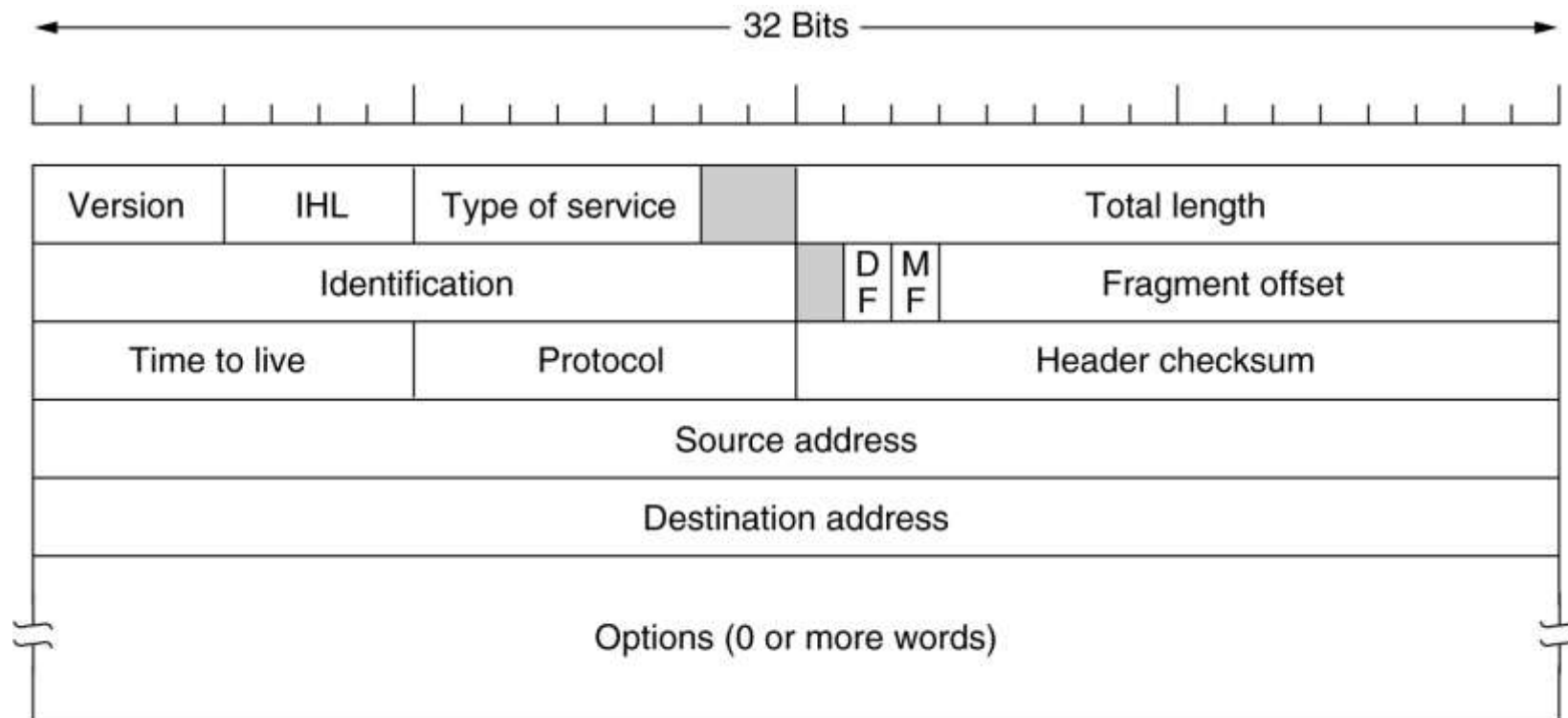
## The Internet: Collections of Subnetworks or ASes



# The Network Layer in the Internet:

## The IPv4 Protocol

An IP datagram consists of a header part and a text part.  
The IPv4 (Internet Protocol) header.



# The Network Layer in the Internet:

## The IPv4 Protocol:

### Header fields

- *Version*: to keep track of which version of the protocol the datagram belongs to.
- *IHL*: to tell how long the header is, in 32-bit words.  $5 \leq \text{IHL} \leq 15$ .
- *Differentiated service*:
  - *Type of service*: 3 for priority, 3 for D, T, and R, and 2 unused.
  - *Differentiated services*: 6 for service class, 2 for congestion.
- *Total length*: the length of header and data. The maximum length is 65,535 bytes.
- *Identification*: the ID of the datagram.

# The Network Layer in the Internet:

## The IPv4 Protocol:

### Header fields

- *DF*: Don't Fragment ( $\geq 576$ )
- *MF*: More Fragment (13)
- *Fragment offset*: to tell where in the current datagram this fragment belongs
- *Time to Live*: a counter used to limit packet lifetimes.
- *Protocol*: which transport process to give this datagram to.  
(<http://www.iana.org/assignments/protocol-numbers>)
- *Header checksum*: to verify the header only
- *Source and destination address*: to indicate the network number and host number.
- Options

# The Network Layer in the Internet:

## IP Header Options

Some of the IP options.

| Option                | Description                                        |
|-----------------------|----------------------------------------------------|
| Security              | Specifies how secret the datagram is               |
| Strict source routing | Gives the complete path to be followed             |
| Loose source routing  | Gives a list of routers not to be missed           |
| Record route          | Makes each router append its IP address            |
| Timestamp             | Makes each router append its address and timestamp |

# The Network Layer in the Internet:

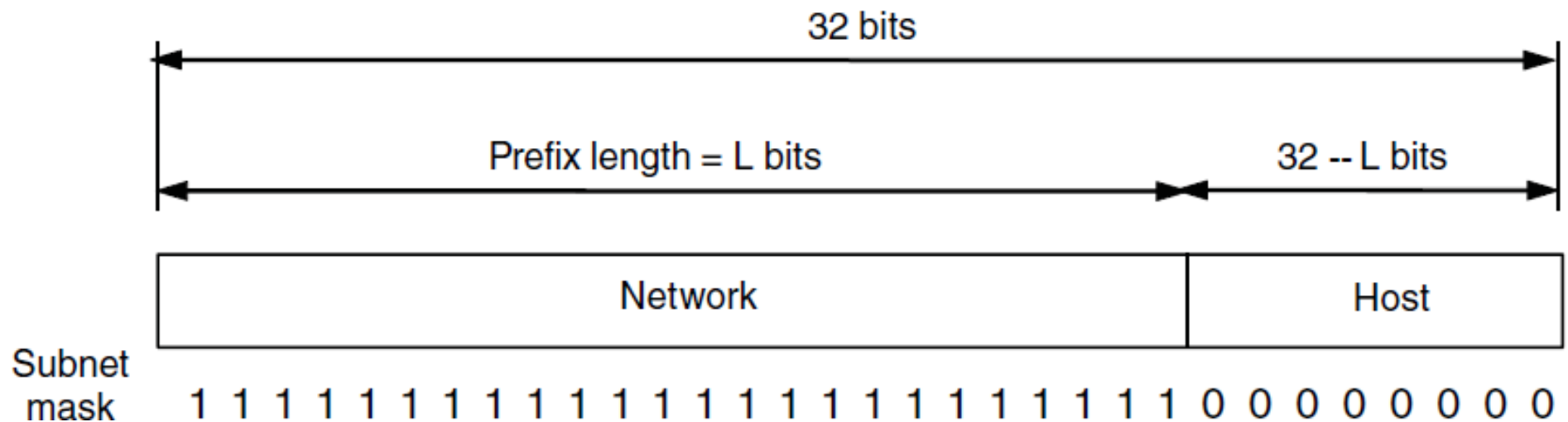
## IP Addresses

- Every Internet interface has an IP address, which encodes its network number and host number. The combination is unique: no two interfaces have the same IP address.
- All IP addresses are 32 bits long and are used in the source address and Destination address fields of IP packets
- IP addressing
  - Prefixes
  - Subnets (division),
  - CIDR(mergement),
  - Classful and Special Addressing
  - NAT

# The Network Layer in the Internet:

## IP Addresses: Prefixes

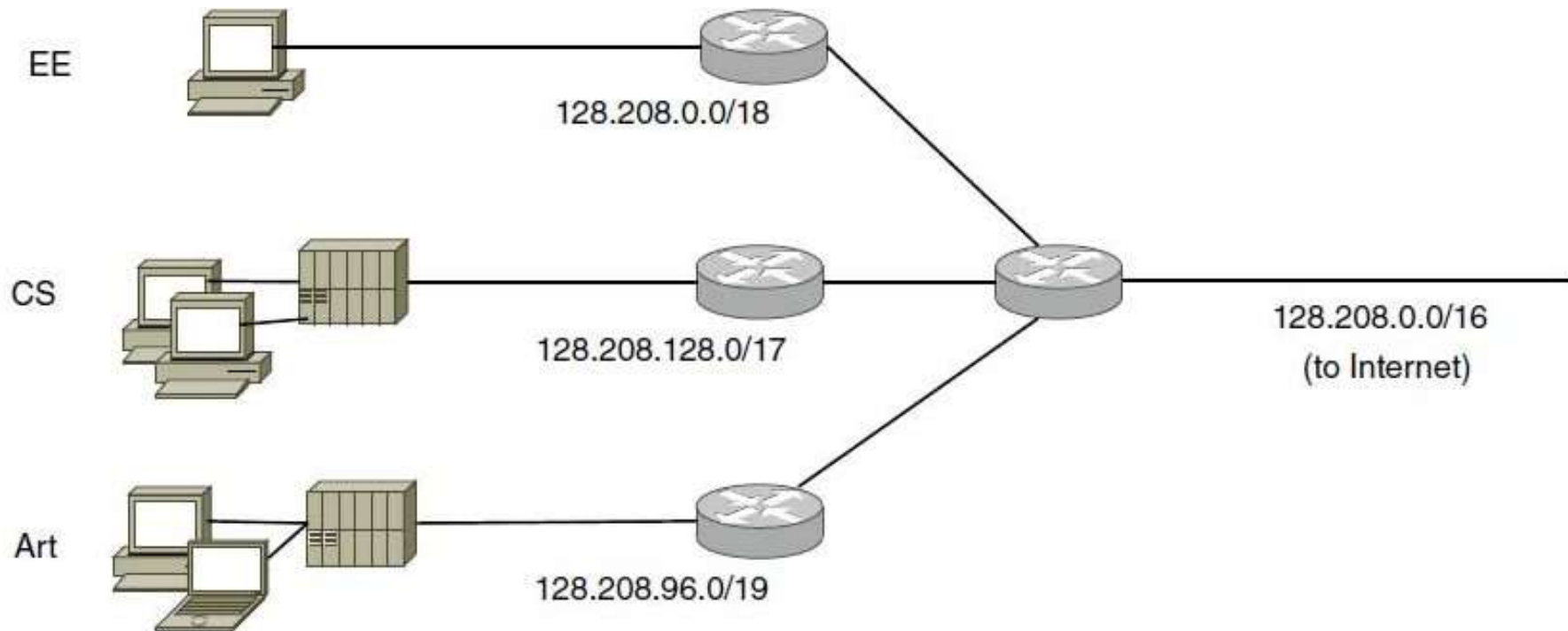
An IP prefix.



# The Network Layer in the Internet:

## IP Addresses: Prefixes

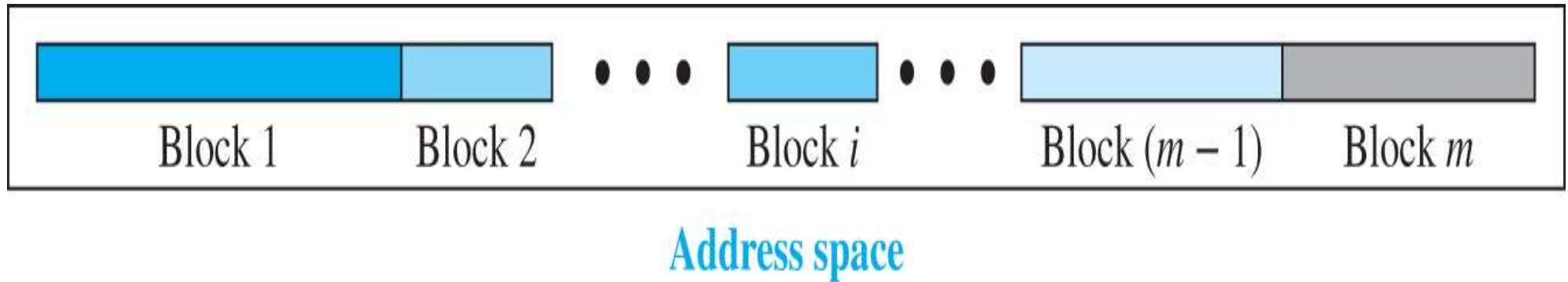
Splitting an IP prefix into separate networks with subnetting.



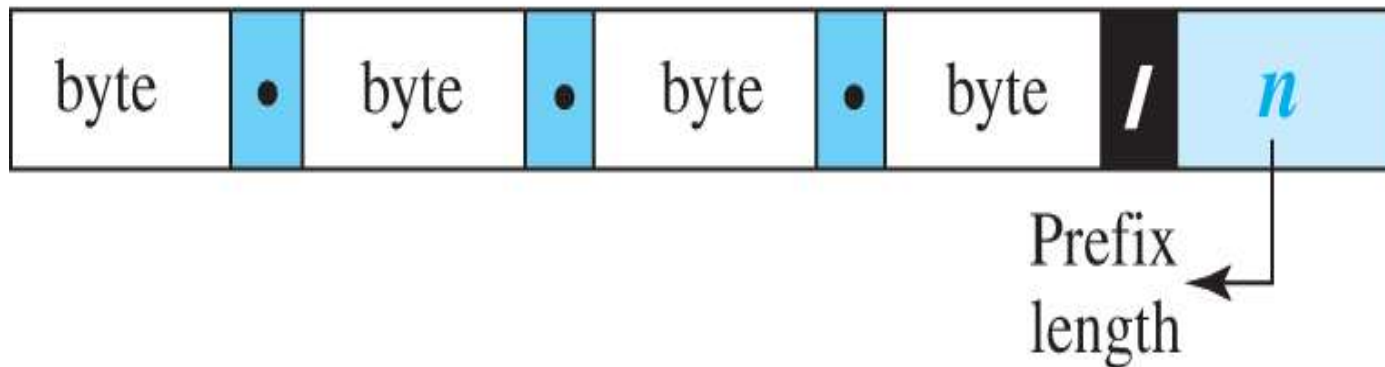
# Classless Addressing

With the growth of the Internet, it was clear that a larger address space was needed as a long-term solution. The larger address space, however, requires that the length of IP addresses also be increased, which means the format of the IP packets needs to be changed. Although the long-range solution has already been devised and is called IPv6, a short-term solution was also devised to use the same address space but to change the distribution of addresses to provide a fair share to each organization. The short-term solution still uses IPv4 addresses, but it is called classless addressing.

Figure 7.7 Variable-length blocks in classless addressing



## Figure 7.8 Slash notation or Classless interdomain routing (CIDR)



### Examples:

12.24.76.8/**8**

23.14.67.92/**12**

220.8.24.255/**25**

## ***Extracting Information from an Address***

- The information from the address blocks can be extracted as the number of addresses, the first address in the block, and the last address.
- To find the first address, we keep the  $n$  leftmost bits and set the  $(32 - n)$  rightmost bits all to 0s.
- To find the last address, we keep the  $n$  leftmost bits and set the  $(32 - n)$  rightmost bits all to 1s.

## Example 7.1

A classless address is given as 167.199.170.82/27. We can find the desired three pieces of information as follows. The number of addresses in the network is  $2^{32-n} = 2^5 = 32$  addresses. The first address can be found by keeping the first 27 bits and changing the rest of the bits to 0s.

|                                  |          |          |          |          |
|----------------------------------|----------|----------|----------|----------|
| Address: 167.199.170.82/27       | 10100111 | 11000111 | 10101010 | 01010010 |
| First address: 167.199.170.64/27 | 10100111 | 11000111 | 10101010 | 01000000 |

The last address can be found by keeping the first 27 bits and changing the rest of the bits to 1s.

|                                 |          |          |          |          |
|---------------------------------|----------|----------|----------|----------|
| Address: 167.199.170.82/27      | 10100111 | 11000111 | 10101010 | 01011111 |
| Last address: 167.199.170.95/27 | 10100111 | 11000111 | 10101010 | 01011111 |

# The Network Layer in the Internet:

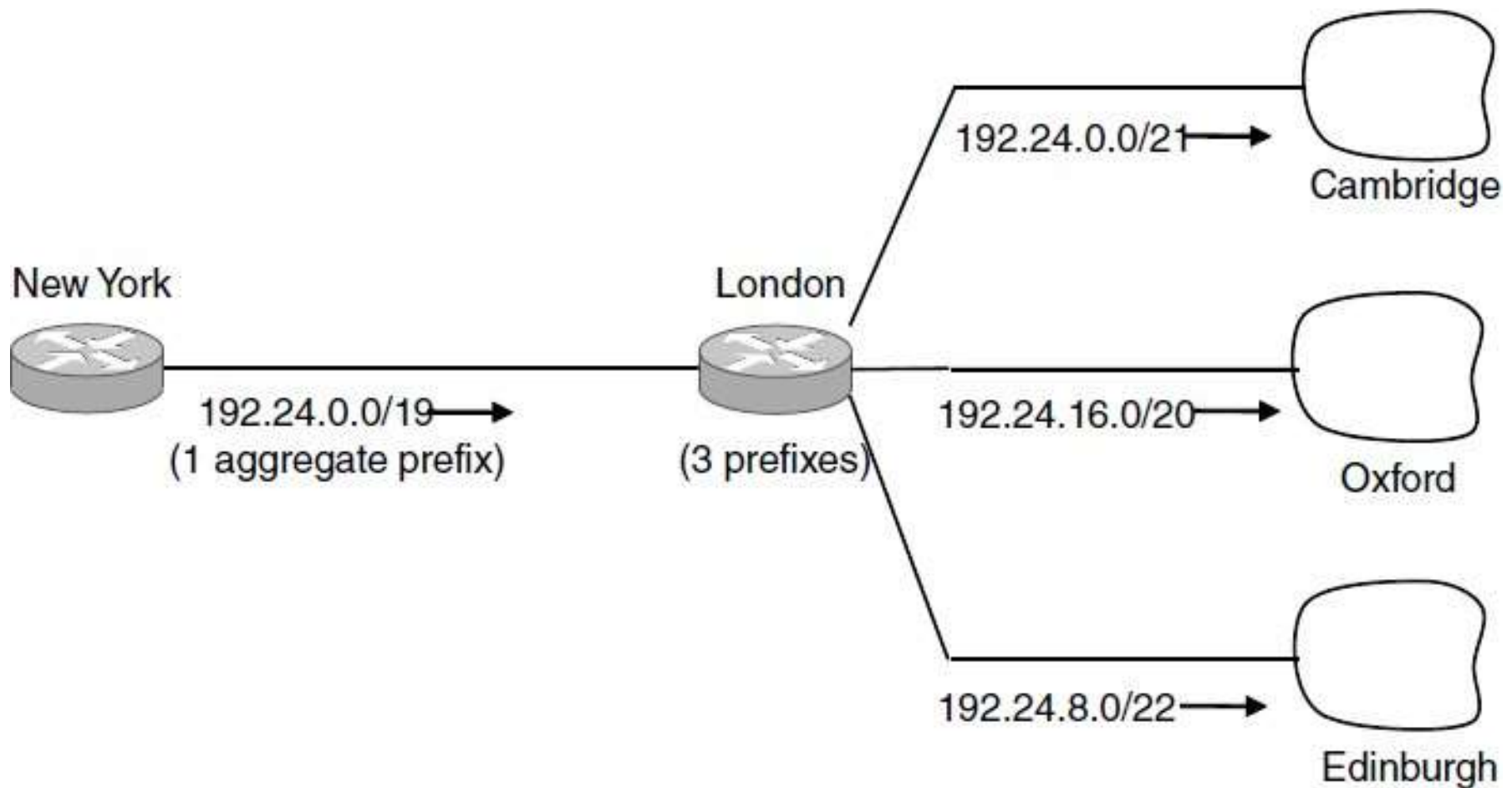
## IP Addresses: CIDR

A set of IP address assignments

| University  | First address | Last address  | How many | Prefix         |
|-------------|---------------|---------------|----------|----------------|
| Cambridge   | 194.24.0.0    | 194.24.7.255  | 2048     | 194.24.0.0/21  |
| Edinburgh   | 194.24.8.0    | 194.24.11.255 | 1024     | 194.24.8.0/22  |
| (Available) | 194.24.12.0   | 194.24.15.255 | 1024     | 194.24.12/22   |
| Oxford      | 194.24.16.0   | 194.24.31.255 | 4096     | 194.24.16.0/20 |

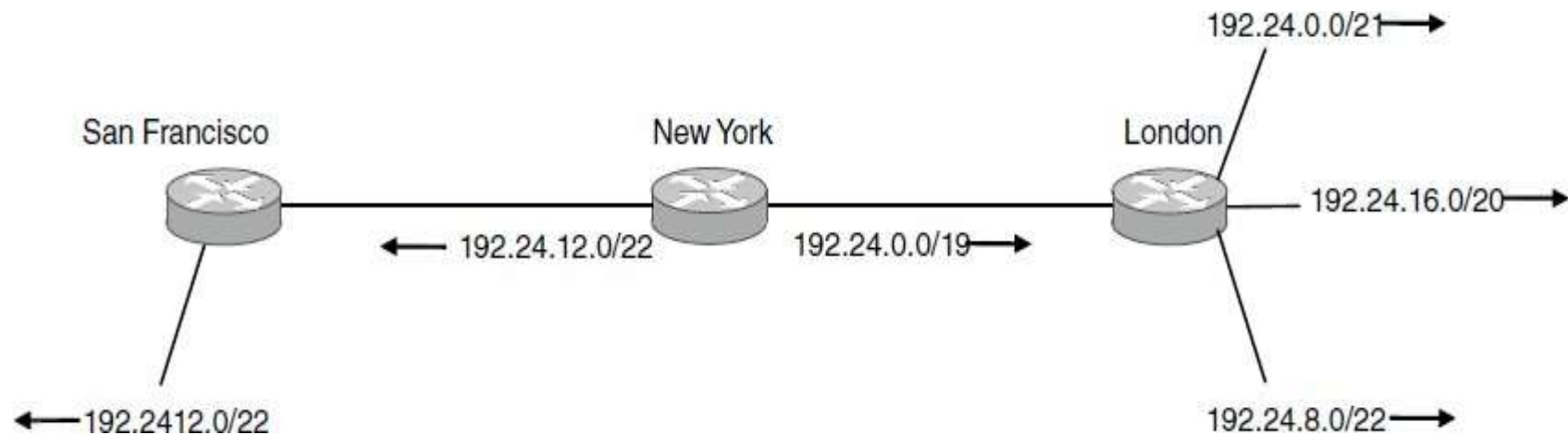
# The Network Layer in the Internet: IP Addresses: CIDR

## Aggregation of IP prefixes



# The Network Layer in the Internet: IP Addresses: CIDR

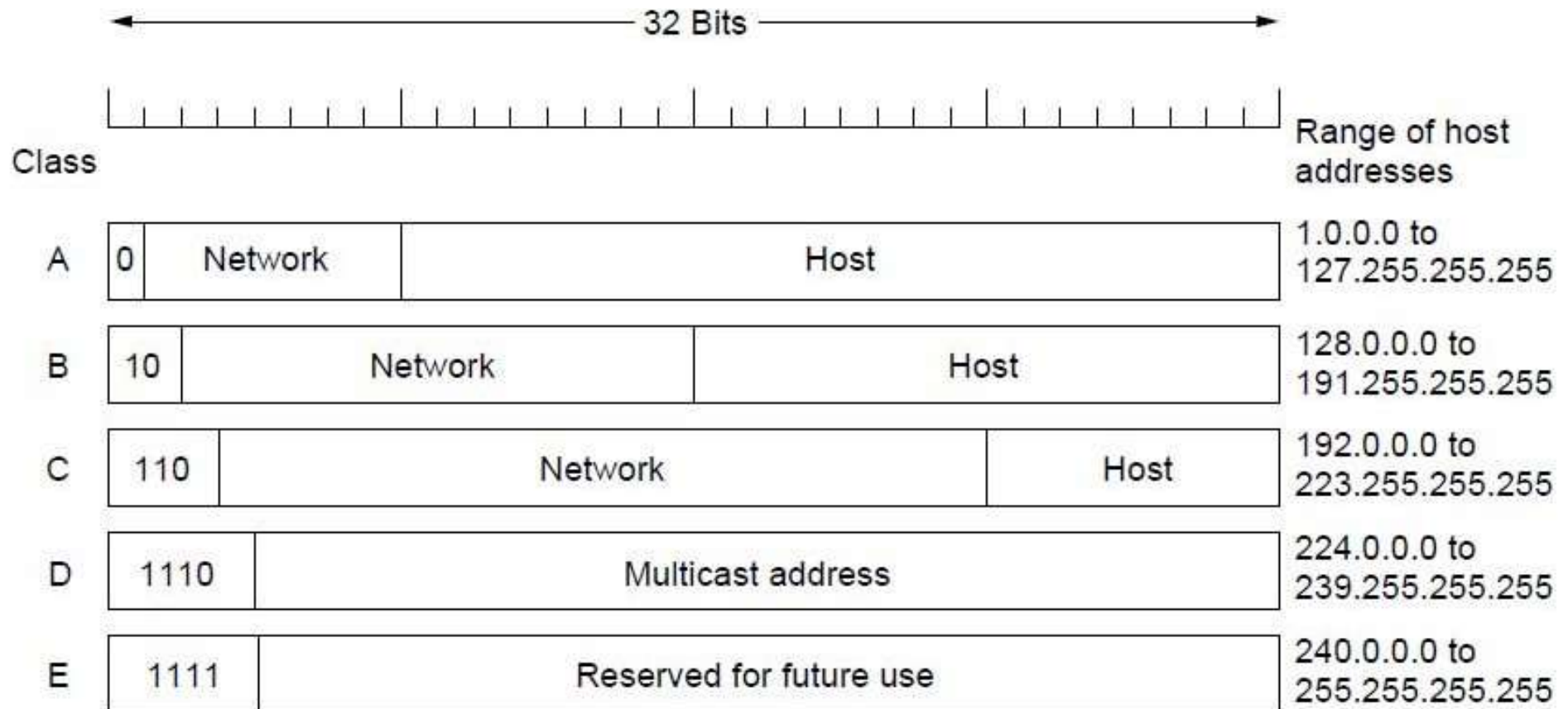
Longest matching prefix routing at the New York router.



# The Network Layer in the Internet:

## IP Addresses: Classful and Special Addressing

### IP address formats



# The Network Layer in the Internet:

## IP Addresses: Classful and Special Addressing

### Special IP addresses

|                                                                 |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  |         |  |  |  |     |  |  |  |      |  |  |  |         |  |  |  |                                |
|-----------------------------------------------------------------|--|--|--|--|--|--|--|------------|--|--|--|--|--|--|--|---------|--|--|--|-----|--|--|--|------|--|--|--|---------|--|--|--|--------------------------------|
| 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  |         |  |  |  |     |  |  |  |      |  |  |  |         |  |  |  | This host                      |
| 0 0                                                             |  |  |  |  |  |  |  | ...        |  |  |  |  |  |  |  | 0 0     |  |  |  |     |  |  |  | Host |  |  |  |         |  |  |  | A host on this network         |
| 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  |         |  |  |  |     |  |  |  |      |  |  |  |         |  |  |  | Broadcast on the local network |
| Network                                                         |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  | 1 1 1 1 |  |  |  | ... |  |  |  |      |  |  |  | 1 1 1 1 |  |  |  | Broadcast on a distant network |
| 127                                                             |  |  |  |  |  |  |  | (Anything) |  |  |  |  |  |  |  |         |  |  |  |     |  |  |  |      |  |  |  |         |  |  |  | Loopback                       |

# The Network Layer in the Internet:

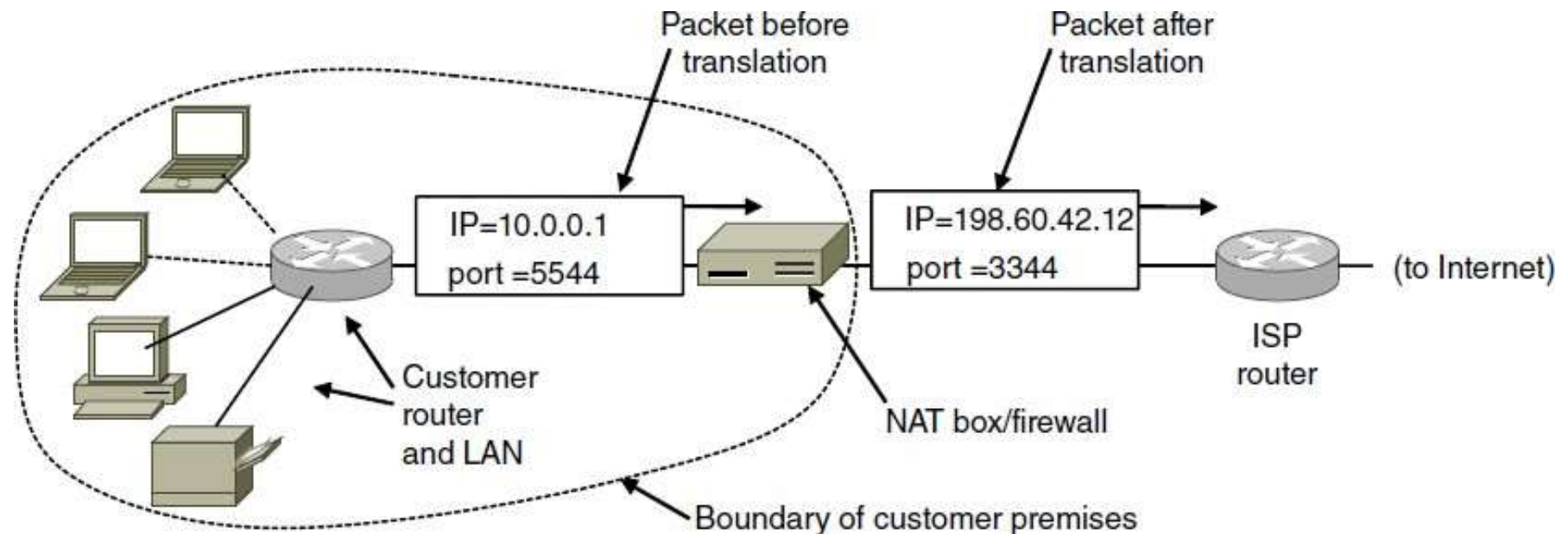
## IP Addresses: NAT(Network Address Translation)

- Addressing
  - Permanent addresses: too few
  - Temporary addresses (DHCP): large users
- Direct addressing (1-level) → indirect addressing (two-level)
- NAT (Network Address Translation)
  - To assign each company a single IP address (or at most, a small number of them) for Internet traffic
  - Within the company, every computer gets a unique IP address (10.x.x.x, 172.16.x.x, 192.168.x.x)

# The Network Layer in the Internet:

## IP Addresses: NAT

Placement and operation of a NAT box.



# The Network Layer in the Internet:

## IP Addresses: NAT

- How NAT works internally
  - (IP add\_src, port\_src)  $\leftrightarrow$  (IP add\_dst, port\_dst)
  - For outgoing packets
    - (10.0.0.1, 5544)  $\rightarrow$  (198.60.42.12, 3344)
    - (198.60.42.12, 3344)  $\rightarrow$  (210.32.32.32, 4433)
    - (198.60.42.12, 3344)  $\leftarrow$  (210.32.32.32, 4433)
    - (10.0.0.1, 5544)  $\leftarrow$  (198.60.42.12, 3344)

# The Network Layer in the Internet: IPv6

## Major goals for IPV6

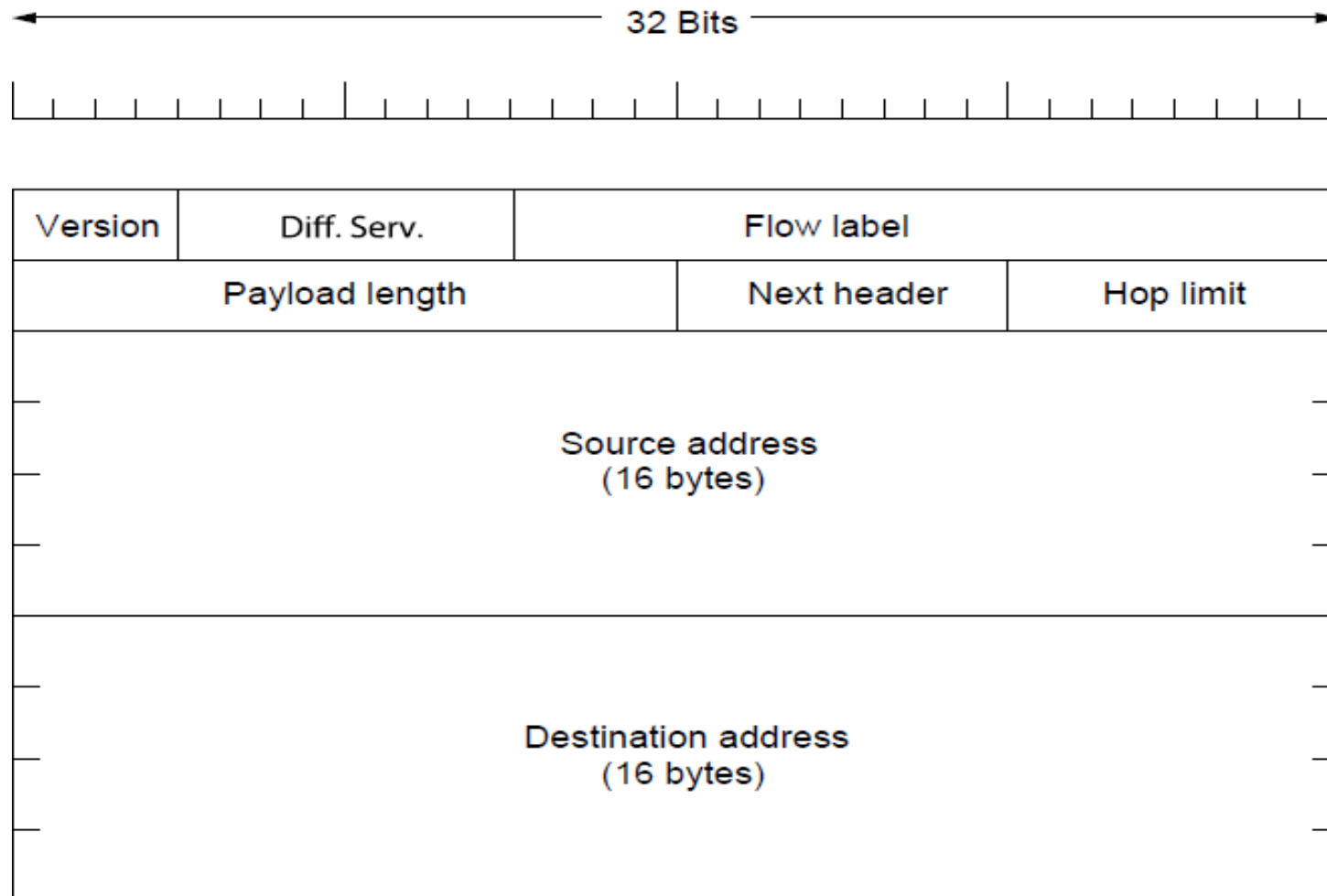
- Support billions of hosts.
- Reduce the size of the routing tables.
- Simplify the protocol, to allow routers to process packets faster.
- Provide better security (authentication and privacy).
- Pay more attention to type of service.
- Aid multicasting by allowing scopes to be specified.
- Make it possible for a host to roam without changing its address.
- Allow the protocol to evolve in the future.
- Permit the old and new protocols to coexist for years

# The Network Layer in the Internet: IPv6

- In 1990, IETF started work on a new version of IP.
- IETF issued a call for proposals and discussion in RFC 1550(IP: Next Generation (IPng) White Paper Solicitation).
- 21 in 1990 → 7 in 1992 → 3 in 1993 → 1
- IPv6 meets the goals fairly well.
  - IPv6 has longer addresses than IPv4. They are 16 bytes long.
  - The simplification of the header.
  - Better support for options.
  - Security
  - More attention has been paid to type of service than in the past.

# The Network Layer in the Internet: IPv6

## The IPv6 fixed header (required).



# The Network Layer in the Internet: IPv6

IPv6 extension headers.

| Extension header           | Description                                |
|----------------------------|--------------------------------------------|
| Hop-by-hop options         | Miscellaneous information for routers      |
| Destination options        | Additional information for the destination |
| Routing                    | Loose list of routers to visit             |
| Fragmentation              | Management of datagram fragments           |
| Authentication             | Verification of the sender's identity      |
| Encrypted security payload | Information about the encrypted contents   |

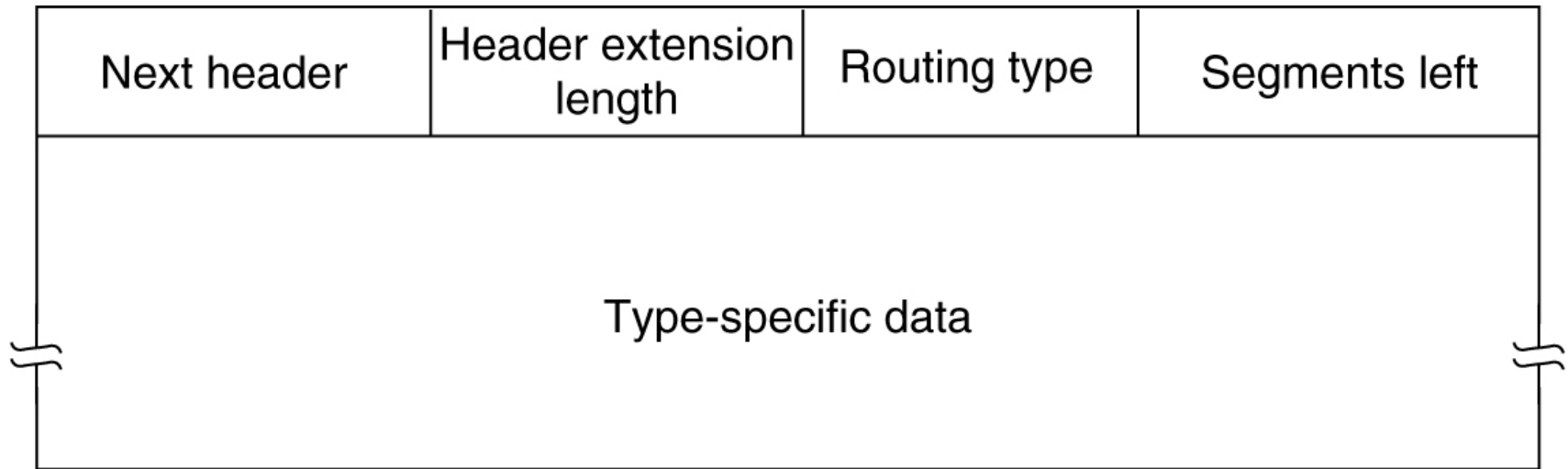
# The Network Layer in the Internet: IPv6

The hop-by-hop extension header for large datagrams (jumbograms).

|                      |   |     |   |
|----------------------|---|-----|---|
| Next header          | 0 | 194 | 4 |
| Jumbo payload length |   |     |   |

# The Network Layer in the Internet: IPv6

IPv6 extension headers.



# The Network Layer in the Internet: IPv6

## Controversies

- Address length: 8 byte, 16 byte, 20 bytes → 16bytes
  - Hop limit: 8 bits or more → 8 bits
  - Maximum packet size: 64 KB or larger → normal 64kB and permit jumbograms.
  - Checksum: needed or not → not needed any more.
  - Mobile support: yes or no → no but.
  - Security: yes or no → no but.
- 
- Huitema, C. 1998. “IPv6: The New Internet Protocol” Prentice-Hall.

# The Network Layer in the Internet:

## Internet Control Protocols: ICMP

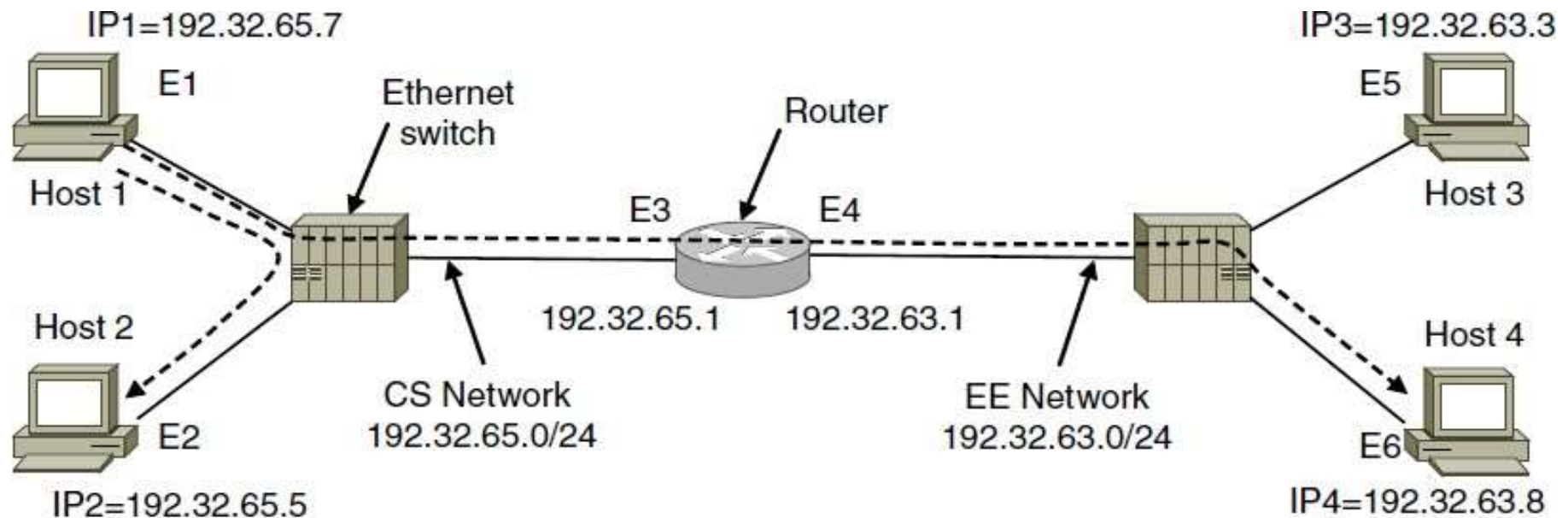
The principal ICMP message types.

| Message type                      | Description                      |
|-----------------------------------|----------------------------------|
| Destination unreachable           | Packet could not be delivered    |
| Time exceeded                     | Time to live field hit 0         |
| Parameter problem                 | Invalid header field             |
| Source quench                     | Choke packet                     |
| Redirect                          | Teach a router about geography   |
| Echo and Echo reply               | Check if a machine is alive      |
| Timestamp request/reply           | Same as Echo, but with timestamp |
| Router advertisement/solicitation | Find a nearby router             |

# The Network Layer in the Internet:

## Internet Control Protocols: ARP

Two switched Ethernet LANs joined by a router



| Frame                  | Source IP | Source Eth. | Destination IP | Destination Eth. |
|------------------------|-----------|-------------|----------------|------------------|
| Host 1 to 2, on CS net | IP1       | E1          | IP2            | E2               |
| Host 1 to 4, on CS net | IP1       | E1          | IP4            | E3               |
| Host 1 to 4, on EE net | IP1       | E4          | IP4            | E6               |

# The Network Layer in the Internet:

## Internet Control Protocols: ARP

How does a user on host 1 send a packet to a user on host 2?

- Find the IP address for host 2 (e.g. DNS)
- Build a packet with 192.31.65.5 in the Destination address field
- Find the destination's Ethernet address: Conf File or ARP
- Build an Ethernet frame addressed to E2 and dump it into the Ethernet.
- The Ethernet board of host 2 detects this frame, recognizes it as a frame for itself, and causes an interrupt.
- The Ethernet driver extracts the IP packet from the payload and passes it to the IP software.

# The Network Layer in the Internet:

## Internet Control Protocols: ARP

How host 1 sends a packet to host 4 (192.31.63.8).

- Host 1 packets the IP packet and sends the frame to E3.
- When the CS router gets the Ethernet frame, it finds the physical address E3 by ARP. It then inserts the packet into the payload field of a frame addressed to E6 and puts it on the net.
- When the Ethernet frame arrives at host 4, the packet is extracted from the frame and passed to the IP software for processing.

# The Network Layer in the Internet:

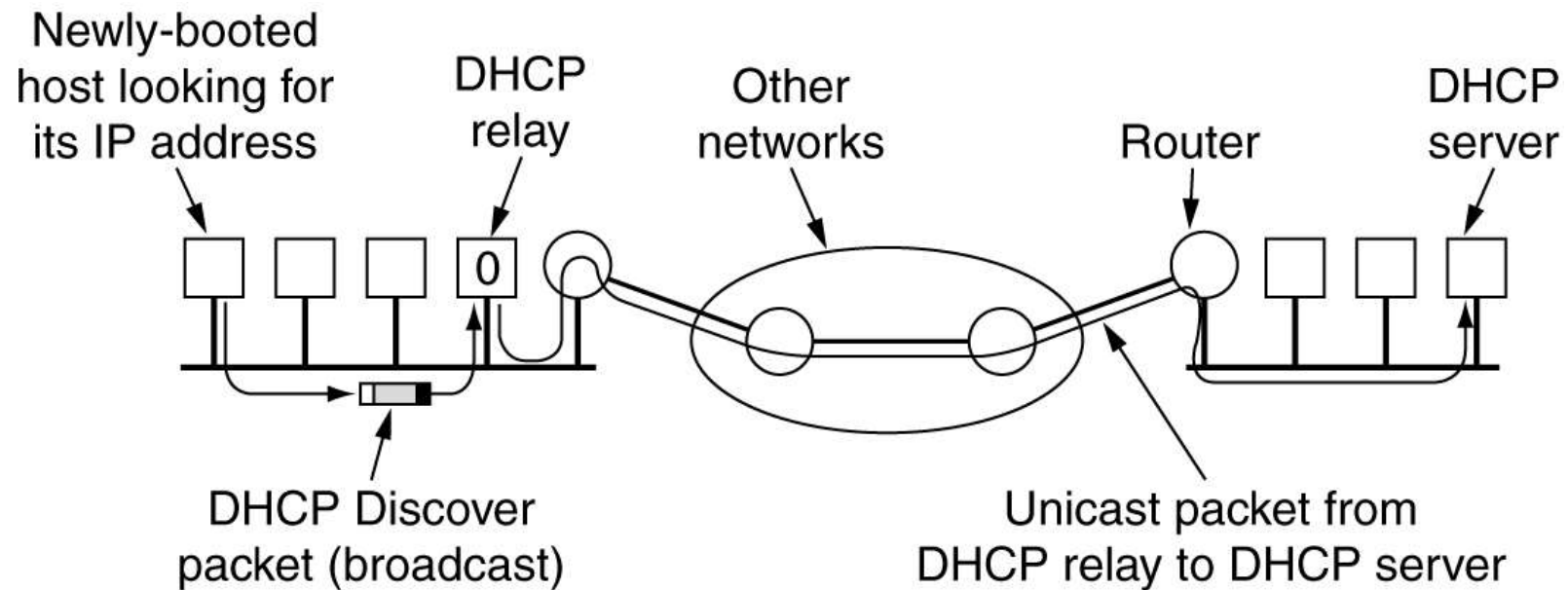
## Internet Control Protocols: DHCP

- ARP (Address Resolution Protocol, )
- RARP (Reverse Address Resolution Protocol): Given a Link address, what is the corresponding IP address?
- BOOTP (Bootstrap Protocol) is better than RARP, can be forwarded by a router.
- DHCP (Dynamic Host Configuration Protocol,) allows both manual IP address assignment and automatic assignment.

# The Network Layer in the Internet:

## Internet Control Protocols: DHCP

### Operation of DHCP.



# The Network Layer in the Internet:

## OSPF – An Interior Gateway Routing Protocol

- The internet is made up of a large number of autonomous systems.
- A routing algorithm within an AS is called an interior gateway protocol
  - Distance vector protocol (Routing Information Protocol)
  - Link state protocol (1979)
  - OSPF (Open Shortest Path First in 1990)
- A routing algorithm between ASes is called an exterior gateway protocol
  - BGP (Border Gateway Protocol)

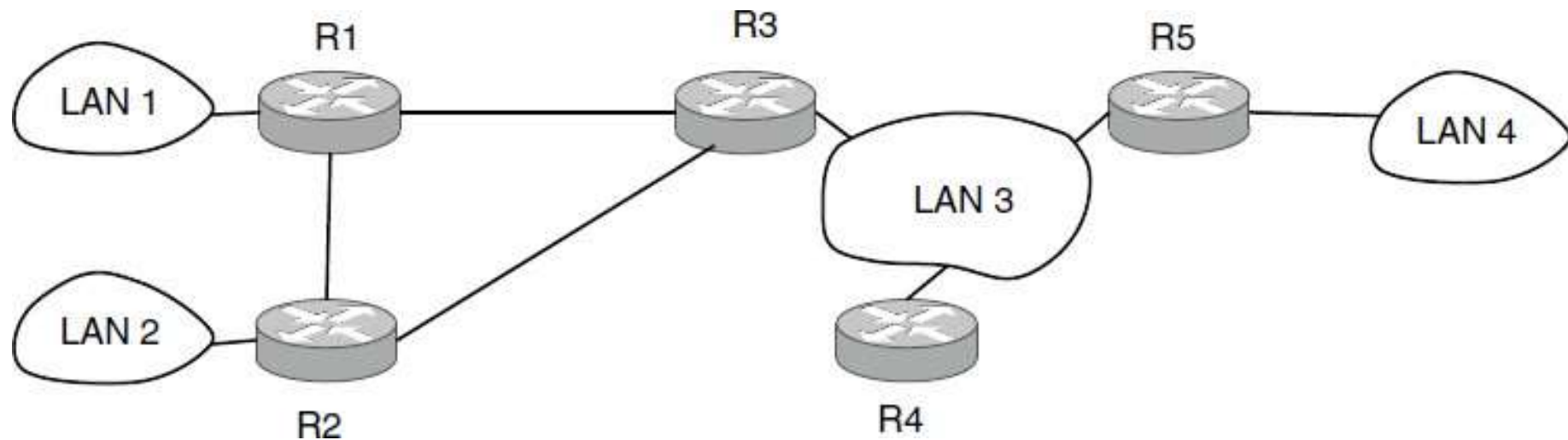
# The Network Layer in the Internet: OSPF

Requirements for the new routing algorithms:

1. To be open, hence the "O" in OSPF.
2. To support a variety of distance metrics.
3. To be a dynamic algorithm.
4. To support routing based on type of service.
5. To do load balancing
6. To support hierarchical systems.
7. To support security
8. To support connection to the Internet via a tunnel

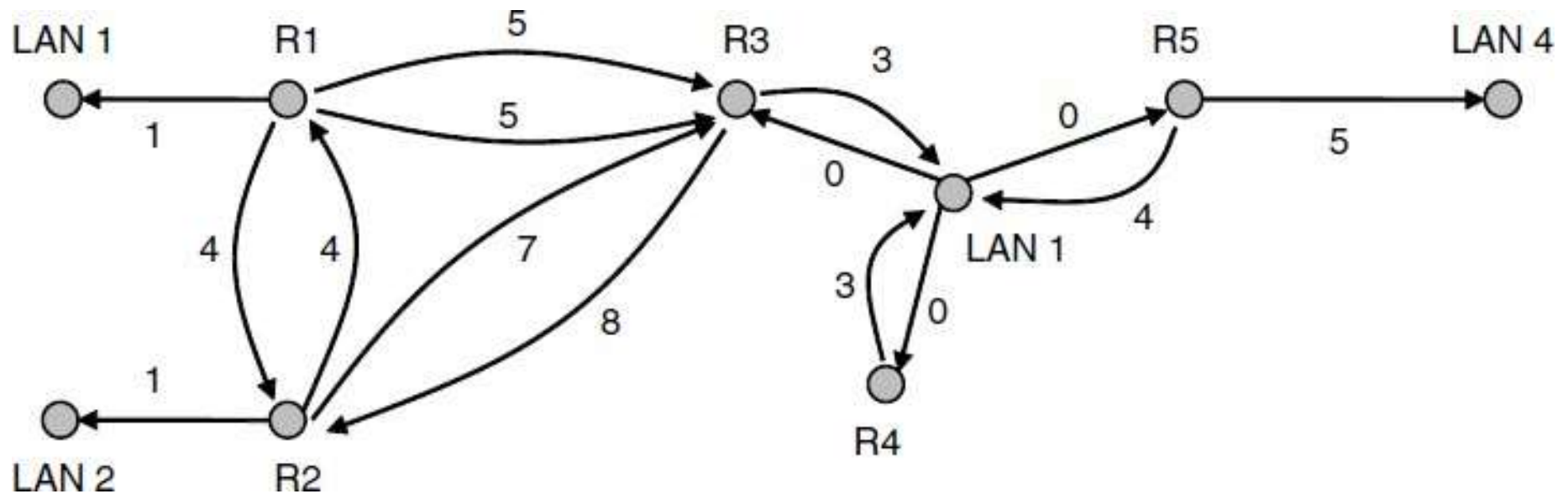
# The Network Layer in the Internet: OSPF

An autonomous system



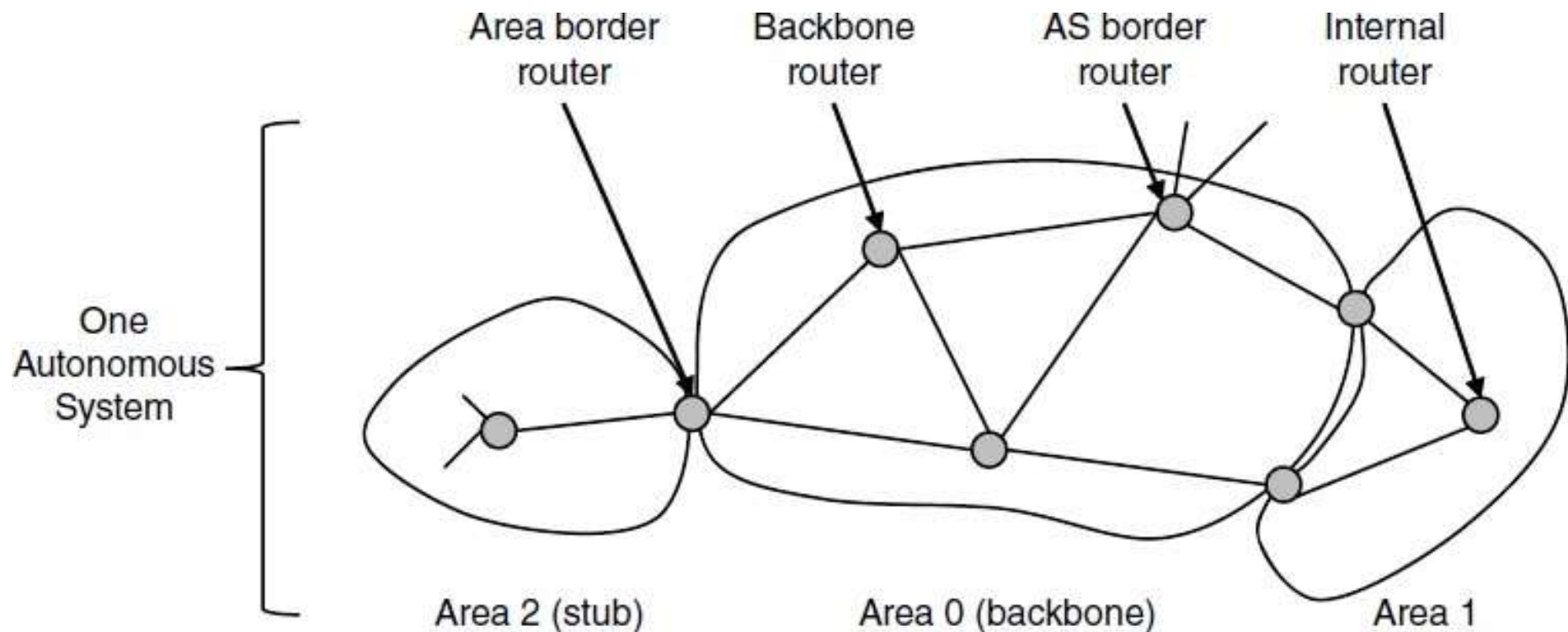
# The Network Layer in the Internet: OSPF

A graph representation of the previous slide.



# The Network Layer in the Internet: OSPF

The relation between ASes, backbones, and areas  
in OSPF.



# The Network Layer in the Internet: OSPF

How a OSPF router works?

- When a router boots, it sends HELLO messages. From the response, each router learns who its neighbors are.
- Adjacent routers exchange information.
  - Each router periodically floods LINK STATE UPDATE messages to each of its adjacent routers. These must be acknowledged (LINK STATE ACK).
  - Either partner can request link state information from the other one using LINK STATE REQUEST messages.
  - Each router constructs the graph for its area(s) and compute the shortest path.

# The Network Layer in the Internet: OSPF

The five types of OSPF messages.

| Message type         | Description                                  |
|----------------------|----------------------------------------------|
| Hello                | Used to discover who the neighbors are       |
| Link state update    | Provides the sender's costs to its neighbors |
| Link state ack       | Acknowledges link state update               |
| Database description | Announces which updates the sender has       |
| Link state request   | Requests information from the partner        |

# The Network Layer in the Internet:

## BGP – The Exterior Gateway Routing Protocol

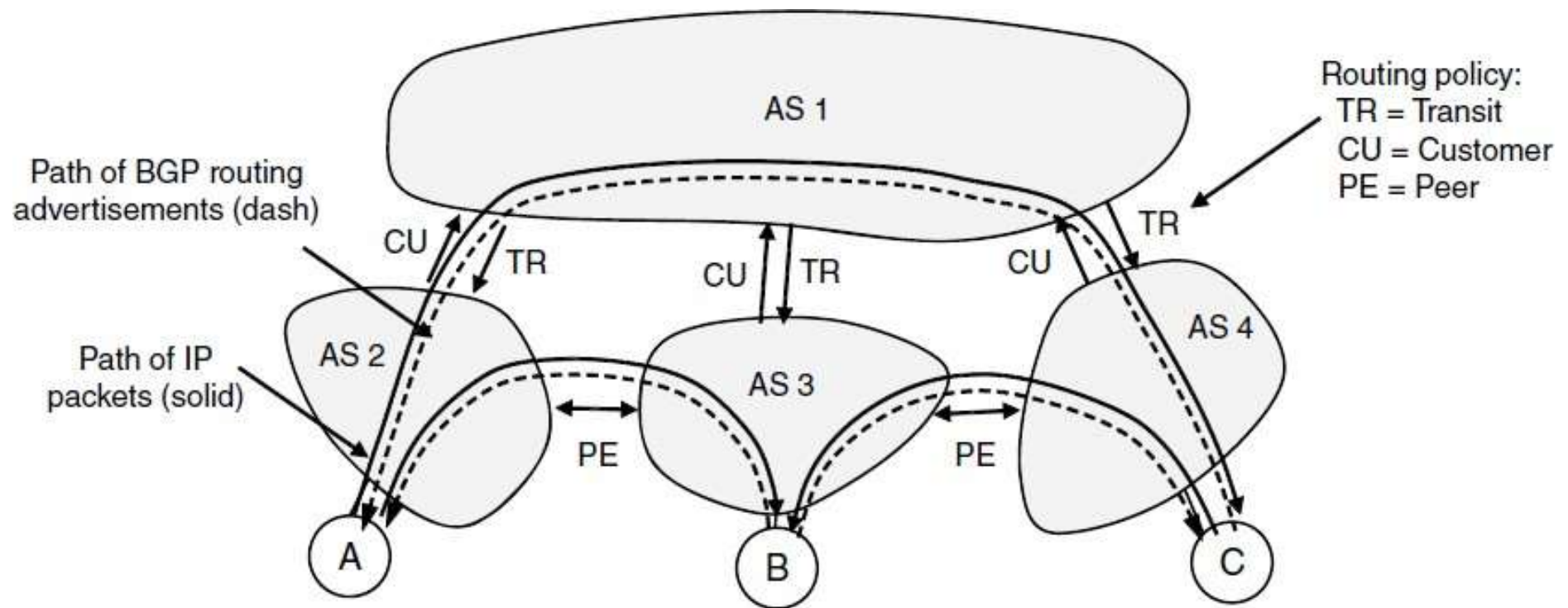
- Exterior gateway protocol routers have to worry about politics a great deal.
- Typical policies involve political, security, or economic considerations. A few examples of routing constraints are
  - 1.No commercial traffic for educat. network
  - 2.Never put Iraq on route starting at Pentagon
  - 3.Choose cheaper network
  - 4.Choose better performing network
  - 5.Don't go from Apple to Google to Apple

# The Network Layer in the Internet: BGP

- BGP routers communicate with each other by establishing TCP connections.
- BGP is fundamentally a distance vector protocol, but quite different from most others such as RIP.
  - Each BGP router keeps track of the path used (instead of maintaining just the cost to each destination)
  - Each BGP router tells its neighbors the exact path it is using (instead of periodically given each neighbor its estimated cost to each possible destination)

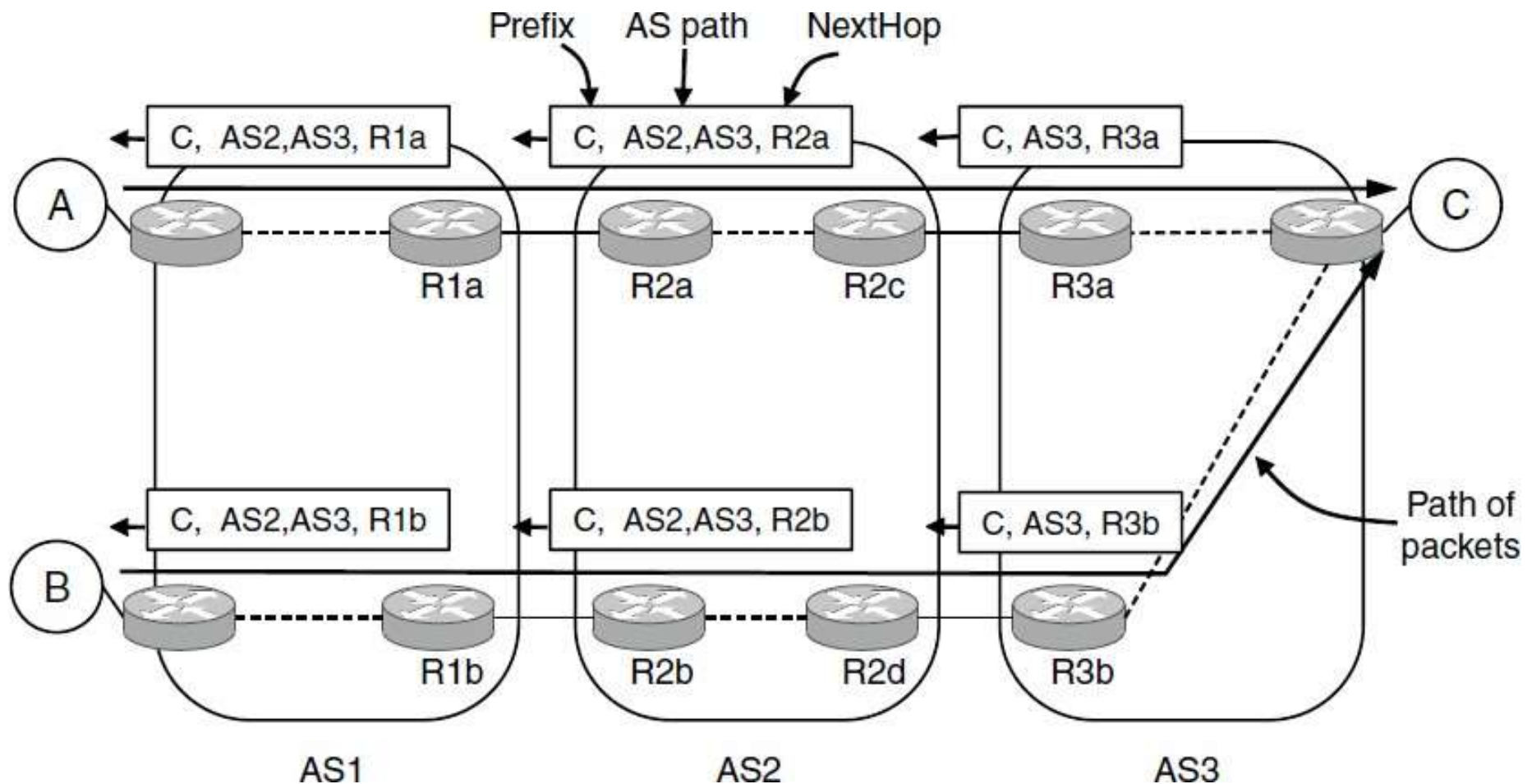
# The Network Layer in the Internet: BGP

## Routing policies between four Autonomous Systems



# The Network Layer in the Internet: BGP

## Propagation of BGP route advertisements



# Homework

- 1
- 6
- 18
- 22
- 31
- 32
- 33
- 42

- 1. Give two example computer applications for which connection-oriented service is appropriate. Now give two examples for which connectionless service is best.

- 6. Consider the network of Fig. 5-12(a). Distance vector routing is used, and the following vectors have just come in to router C: *from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4).* The cost of the links from C to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the cost.

- 18. A token bucket scheme is used for traffic shaping. A new token is put into the bucket every 5  $\mu$ sec. Each token is good for one short packet, which contains 48 bytes of data. What is the maximum sustainable data rate?

- 22. Consider the user of differentiated services with expedited forwarding. Is there a guarantee that expedited packets experience a shorter delay than regular packets? Why or why not?

- 31. A router has just received the following new IP addresses: 57.6.96.0/21, 57.6.104.0/21, 57.6.112.0/21, and 57.6.120.0/21. If all of them use the same outgoing line, can they be aggregated? If so, to what? If not, why not?

- 32. The set of IP addresses from 29.18.0.0 to 29.18.127.255 has been aggregated to 29.18.0.0/17. However, there is a gap of 1024 unassigned addresses from 29.18.60.0 to 29.18.63.255 that are now suddenly assigned to a host using a different outgoing line. Is it now necessary to split up the aggregate address into its constituent blocks, add the new block to the table, and then see if any reaggregation is possible? If not, what can be done instead?

- 33. A router has the following (CIDR) entries in its routing table:

Address/mask Next hop

135.46.56.0/22 Interface 0

135.46.60.0/22 Interface 1

192.53.40.0/23 Router 1

default Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives?

(a) 135.46.63.10

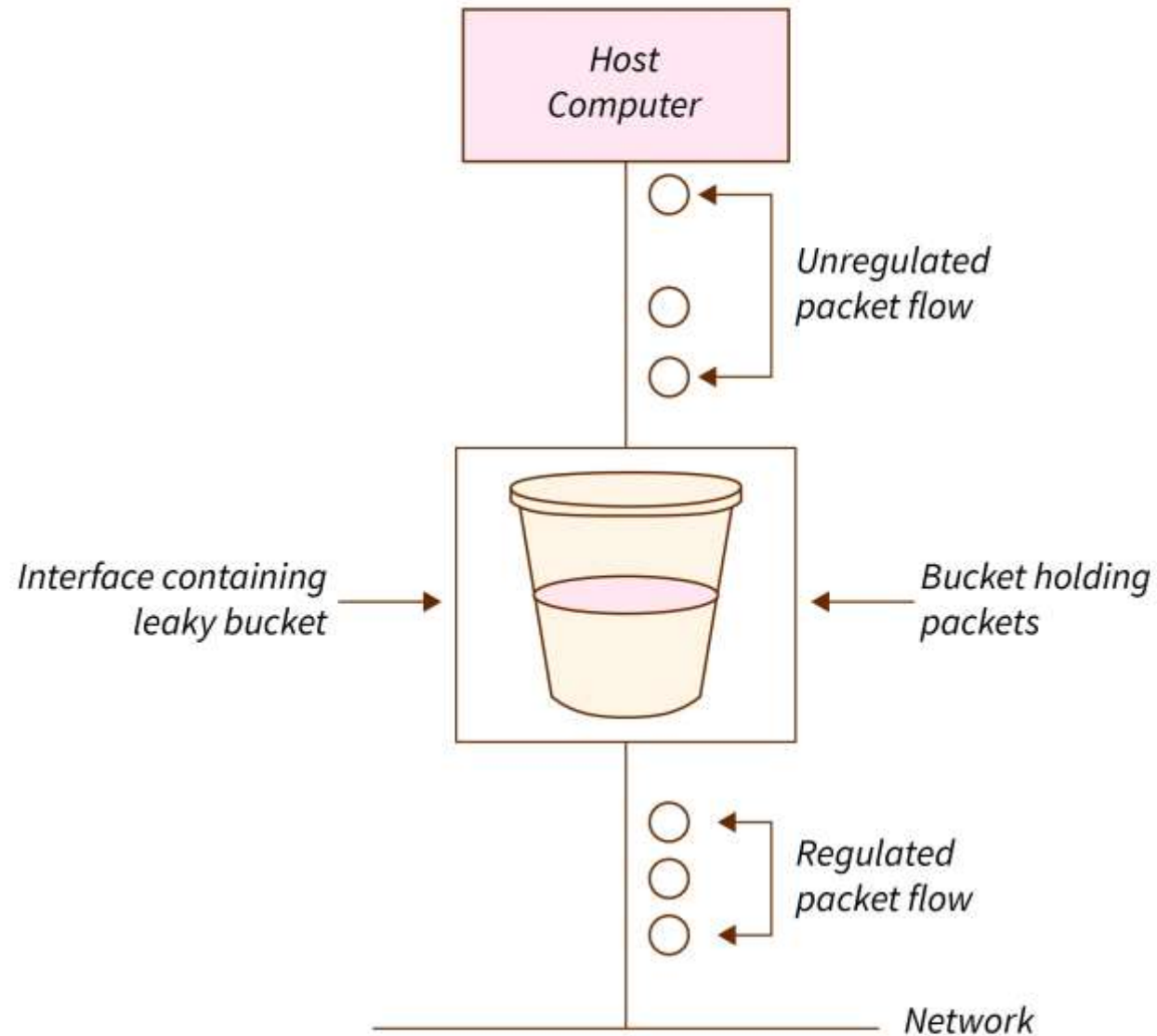
(b) 135.46.57.14

(c) 135.46.52.2

(d) 192.53.40.7

(e) 192.53.56.7

- 42. When the IPv6 protocol is introduced, does the ARP protocol have to be changed? If so, are the changes conceptual or technical?



# Pseudo Code

- Step - 1 :**

Initialize buffer size and leak rate.

- Step - 2 :**

At clock tick, Initialize  $n$  as the leak rate.

- Step - 3 :**

If the  $n$  is greater than the size of the packet at front of the queue, send the packet into the network.

- Step - 4 :**

Decrement  $n$  by the size of the packet.

- Step - 5 :**

Repeat Step 3 and Step 4 until  $n$  is less than the size of the packet at the front of the queue or the bucket is empty. No other packet can be transmitted till the next clock tick.

- Step - 6 :**

Go to Step 2.